GLUCOSE OXIDATION AND SHUTTLES

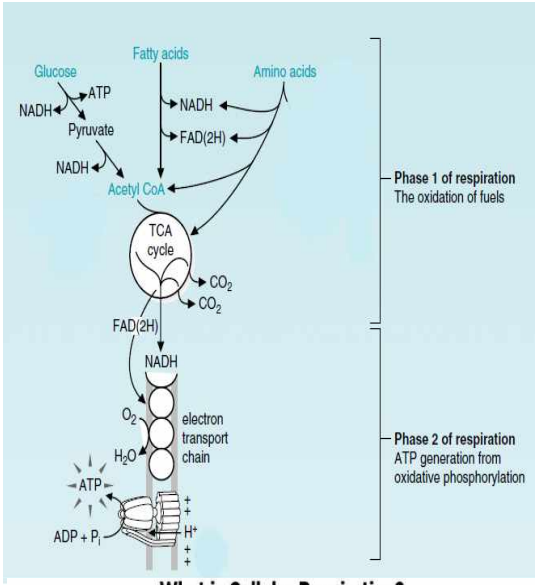
- ✓ All physiologic processes in living cells require energy transformation. Cells convert the chemical bond energy in foods into other forms

CELLULAR RESPIRATION:

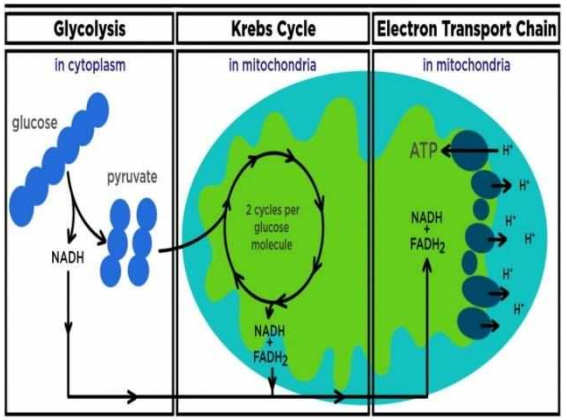
- ✓ The process of using **O2** and energy-derived from **oxidizing fuels** (CHO, lipids, protein) to generate **ATP**.
- ✓ We breathe principally because Cells require O2 to generate ATP from the oxidation of fuels to CO2 and water.
- ✓ Cellular respiration uses over 90% of the O2 we inhale.

These energy transformations can be divided into three principal phases:

Phase 1 Oxidation of fuels (carbohydrates, lipids and proteins).	1-Oxidation of (glucose, fatty acids, and many amino acids) converge on the generation of **acetyl CoA** 2-**complete oxidation** of the **acetyl group** to **CO2** occurs in the **tricarboxylic acid** (TCA) cycle, which collects the energy mostly as **NADH and FADH2.**
Phase 2 of cellular respiration (synthesis of ATP)	1-The **energy derived from fuel oxidation(NADH and FADH2)** is converted to the high-energy phosphate bonds of **ATP** by the process of **oxidative phosphorylation** Electrons are transferred from NADH and FADH2 to O2 by the **electron transport chain** (ETC, located in the inner mitochondrial membrane). These series of reactions are coupled with the synthesis of ATP
Phase 3 of cellular respiration (consumption of ATP)	The high-energy phosphate bonds of **ATP** are used for processes such as **muscle contraction** (mechanical work), **maintaining low intracellular Na concentrations** (transport work), synthesis of larger molecules such as DNA in **anabolic pathways** (biosynthetic work), or **detoxification** (biochemical work). -ATP is either directly or indirectly hydrolyzed to ADP+ inorganic phosphate (Pi), or AMP + pyrophosphate (PPi).



What is Cellular Respiration?



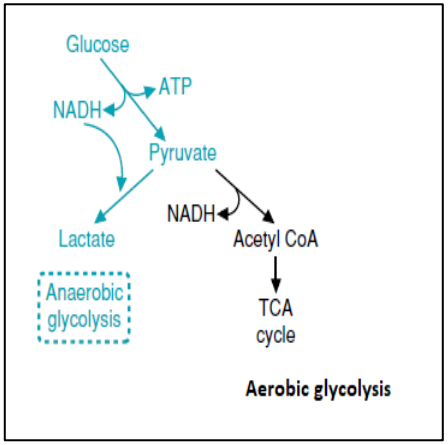
Mitochondria is the site for cellular respiration

- Almost all of the enzymes for the **TCA cycle** and oxidation of fatty acids located in the **mitochondrial matrix (phase 1)**
- The components of the **ETC** are located in the **inner mitochondrial membrane (phase 2)**.

Examples of cellular respiration

❖ **Glucose is a universal fuel used to generate ATP in every cell type in the body**

In glycolysis, **1** mole of **glucose** is oxidized to **2 pyruvate** +**2 NADH** by cytosolic enzymes + **2 ATP** by **substrate-level phosphorylation** from high-energy metabolic intermediates**.**



In anaerobic glycolysis

- **NADH** is reoxidized by conversion of **pyruvate to lactate**, enters the blood.
- Although **anaerobic glycolysis** has a **low ATP yield**, it is important for tissues that **lack of mitochondria** (e.g.,R.B.Cs, the kidney medulla), or **tissues experiencing diminished blood flow** (exercising muscle)

❖ **Fate of the end products of aerobic glycolysis (pyruvate and NADH)**

1- **Pyruvate:** In presence of **O2 and mitochondria**, pyruvate is transported to **mitochondria** and undergoes **oxidative decarboxylation to acetyl CoA** which is the major fuel of the **TCA** cycle.

2- **Cytosolic NADH:** is reoxidized via the **electron transport chain,** but Cytosolic NADH **cannot directly pass** through **inner mitochondrial** membrane to reach the **electron transport chain.**

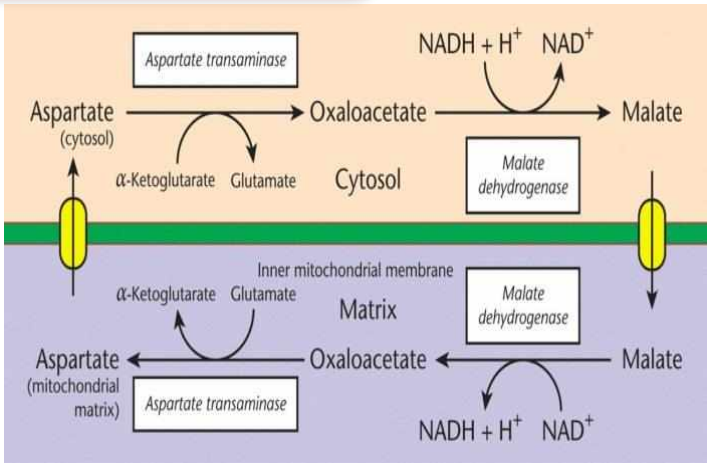
- **NADH is indirectly transferred to the inside of the mitochondria by what is called the shuttle mechanisms**

A-Malate aspartate shuttle

B-Glycerolphosphate shuttle

- Many tissues contain both the glycerophosphate shuttle and the malate–aspartate shuttle.

(1)THE MALATE-ASPARTATE SHUTTLE (MALATE SHUTTLE)



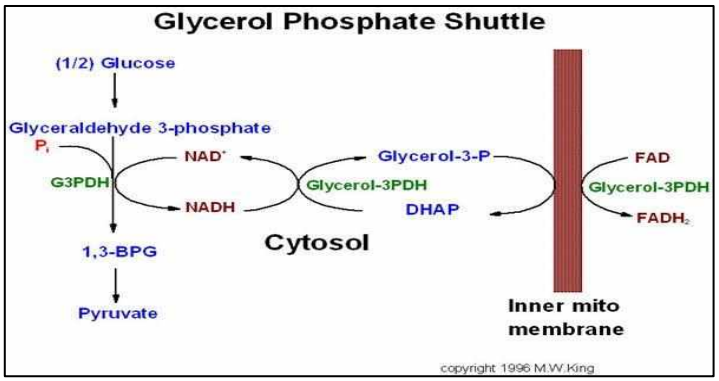
- The cytosolic **malate dehydrogenase (MDH)** enzyme transfers electrons from NADH to cytosolic **Oxaloacetate** ⟶ **Malate.**
- **Malate** is transported across the inner mitochondrial membrane by a specific translocase.
- In the mitochondrial matrix, **Malate** is oxidized back to ⟹ **Oxaloacetate** by **mitochondrial malate dehydrogenase**, and **NADH is generated.**

-This NADH can donate electrons to the **electron transport chain** with generation of approximately **3 moles of ATP per mole of NADH**.

- The newly formed oxaloacetate cannot pass back through the inner mitochondrial membrane under physiologic conditions.

-Therefore, oxaloacetate is converted by **Aspartate transaminase (AST)** into **aspartate**. Then, transported out to the cytosol using an **aspartate/glutamate exchange translocase**

(2) THE GLYCEROLPHOSPHATE SHUTTLE:



- The glycerolphosphate shuttle is the ***major shuttle*** in most tissues.
- Cytoplasmic **glycerol 3-phosphate dehydrogenase** transfers electrons from NADH to **DHAP** to form **glycerol 3-phosphate**
- **Glycerol 3-phosphate** then diffuses to the inner mitochondrial membrane, where the electrons are donated to **FAD forming FADH2.**
- FADH2 by using the **ETC** results in the synthesis of **approximately 2 ATP**.
- Dihydroxyacetone phosphate returns to the cytosol to continue the shuttle.

OXIDATIVE DECARBOXYLATION OF PYRUVATE BY PYRUVATE DEHYDROGENASE COMPLEX (PDH) COMPLEX

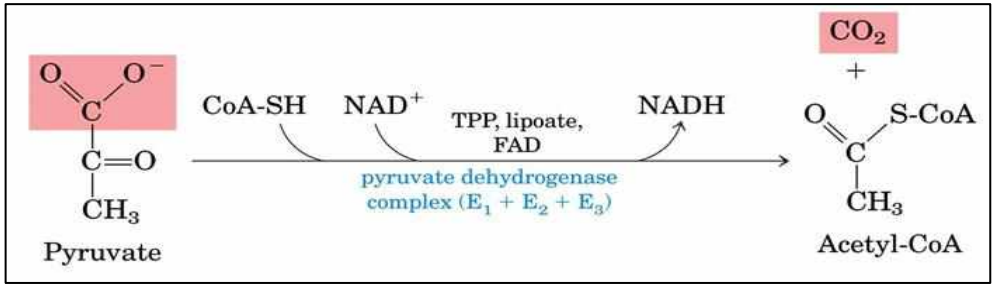
- **Pyruvate** is transported to **the inside of the mitochondria** to be completely oxidized in the **TCA cycle**
- **Within the mitochondria**, pyruvate undergoes **oxidative decarboxylation** catalyzed by the enzyme **pyruvate dehydrogenase complex (PDH)** to form **acetyl CoA**

✓ **The complex contains three enzymes:**

- Pyruvate dehydrogenase, dihyrolipoyl transacetylase and dihydrolipoyl dehydrogenase

✓ **PDH complex needs five coenzymes:**

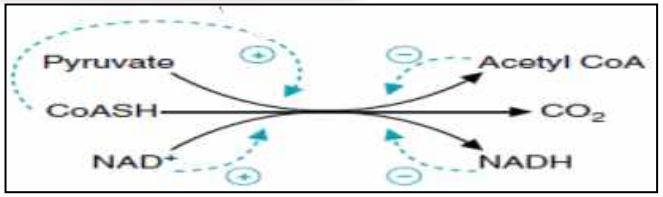
1-Thiamin pyrophosphate (TPP)
2-Lipoic acid
3-COASH
4-FAD
5-NAD+



- Now, the **acetyl CoA** formed from pyruvate is ready to be completely oxidized in **the TCA cycle**.

REGULATION OF PYRUVATE DEHYDROGENASE COMPLEX: مهم

1- Allosteric regulation



PDH is **inhibited** by its products, **acetyl CoA and NADH**.
stimulated by **pyruvate and NAD**

2- Covalent Modification:

✓ PDH is **activated** through **Dephosphorylation** by **pyruvate dehydrogenase phosphatase**
✓ PDH is **inhibited** through **phosphorylation** by **pyruvate dehydrogenase kinase (PDH kinase)**.

PDH KINASE:

PDH kinase is, itself, **inhibited** allosterically by **ADP** and **pyruvate.** (Thus, when rapid ATP utilization results in an **increase of ADP**, or when activation of glycolysis **increases pyruvate levels**, **PDH kinase is inhibited**, and **PDH remains in the active form (non-phosphorylated form).** i.e. (Inhibition of inhibitor) ➡ stimulation

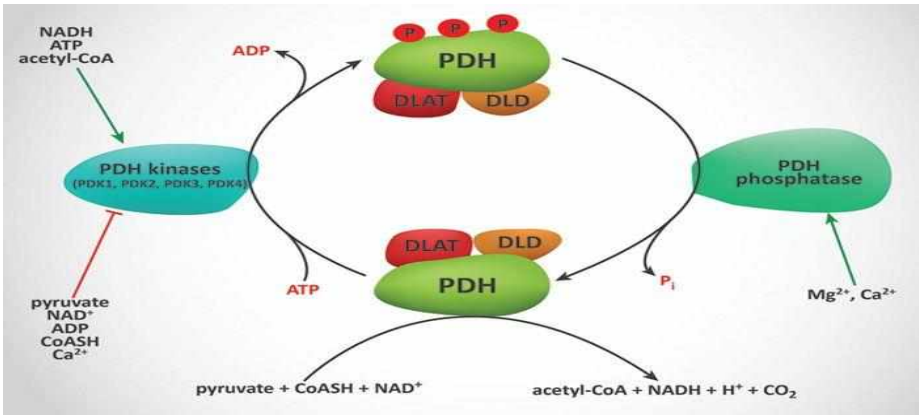
PDH kinase is **activated** by Both **acetyl CoA** and **NADH** which results in **phosphorylation of PDH and inactivation**.

i.e. stimulation of inhibitor ➡ inhibition

PDH PHOSPHATASE:

- ✓ **Requires calcium for its full activity.**

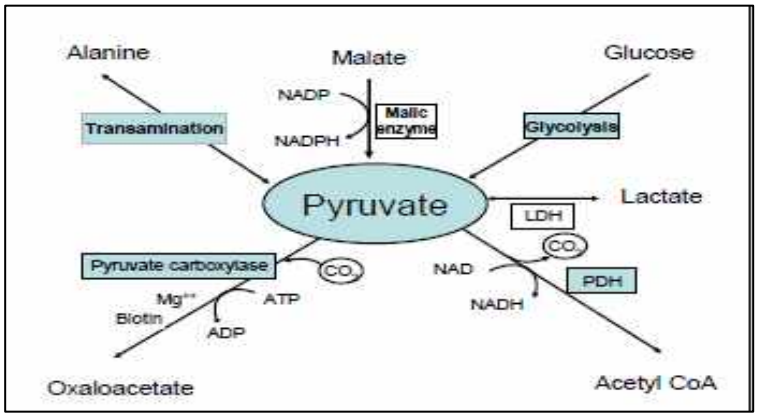
(In the heart, increased intramitochondrial Ca2+ during rapid contraction activates the phosphatase, thereby increasing the amount of active, non-phosphorylated PDH).



N.B **Insulin** favor **the active, dephosphorylated** form while **glucagon** favors the **phosphorylated, inactive** form

Other Fates of Pyruvate:

- ✓ Pyruvate is a central molecule in metabolism. It has several metabolic fates
- ✓ **N.B.** *Pyruvate carboxylase, malic and PDH catalyze irreversible reactions, while lactate dehydrogenase (LDH) and transamination reactions are reversible*



ABNORMALITIES OF THE PDH

may be due to

1. **Congenital deficiency of the enzyme** (which is rare). The defect may be in one or more of the components of the enzyme complex.

2. **Dietary deficiency of thiamin pyrophosphate (TPP)** in populations that use **white polished rice** as a main source of carbohydrates.

- This explains the symptoms and signs noted in patients suffering from **Beriberi**.

3. **Alcohol addiction** is accompanied by nutritional deprivation of **thiamin pyrophosphate** because of impaired intestinal absorption of thiamin. Some individuals with alcoholism develop **Wernicke-Korsakoff syndrome** because brain depends on glucose as a preferred fuel, so it is a prominent tissue where these metabolic defects manifest themselves.

4. **Arsenite:** Inhibits enzymes require **lipoic acid** as a **cofactor as PDH and α- ketoglutarate dehydrogenase complex** (a TCA cycle enzyme). When arsenite binds to lipoic acid in PDH, pyruvate and consequently lactate accumulate.

LACTIC ACIDOSIS:

abnormalities of the PDH lead to accumulation of lactate.
Blood lactate accumulation results in lowering of blood pH (a serious condition).

KREBS CYCLE
(TRICARBOXYLIC ACID CYCLE, TCA)

- ✓ **Krebs cycle** after **Sir Hans Krebs**.
- ✓ Called **citric acid cycle** because citrate was the product of the first reaction
- ✓ The most common name is **TCA cycle**, due to involvement of tricarboxylic acids (citrate and isocitrate).

- The TCA cycle accounts for over **Two thirds of the ATP** generated from fuel oxidation.
  - ❖ **It is the common final catabolic pathway,**

oxidation of fatty acids

glucose
amino acids → **Acetyl CoA** → **substrate for the TCA**
acetate
ketone bodies

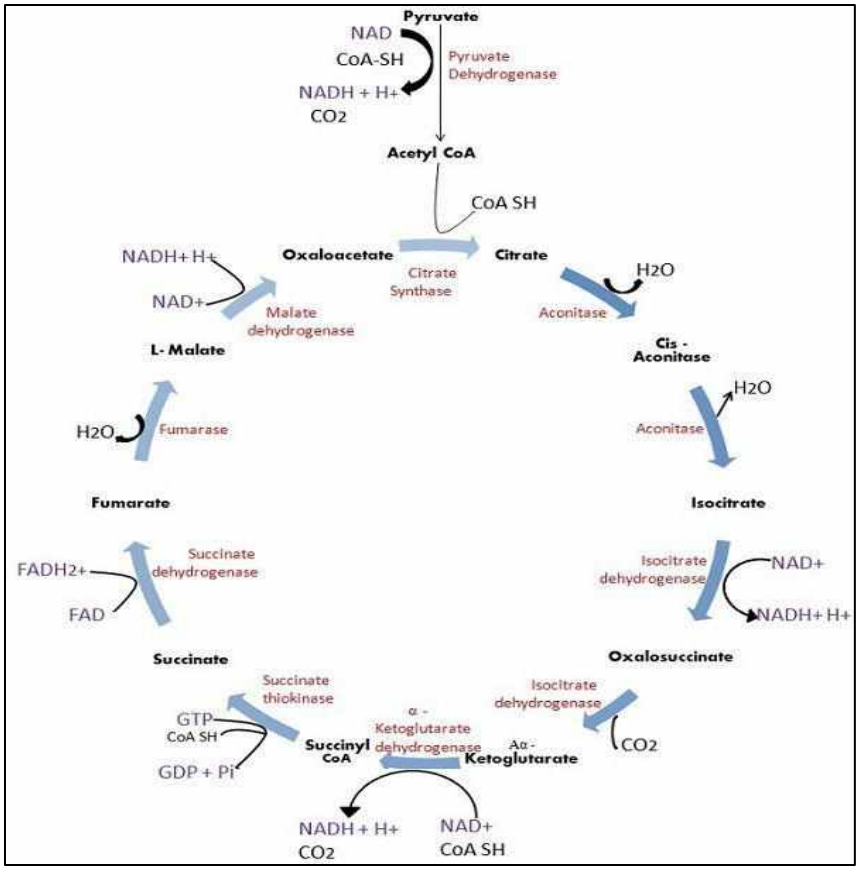
all generate

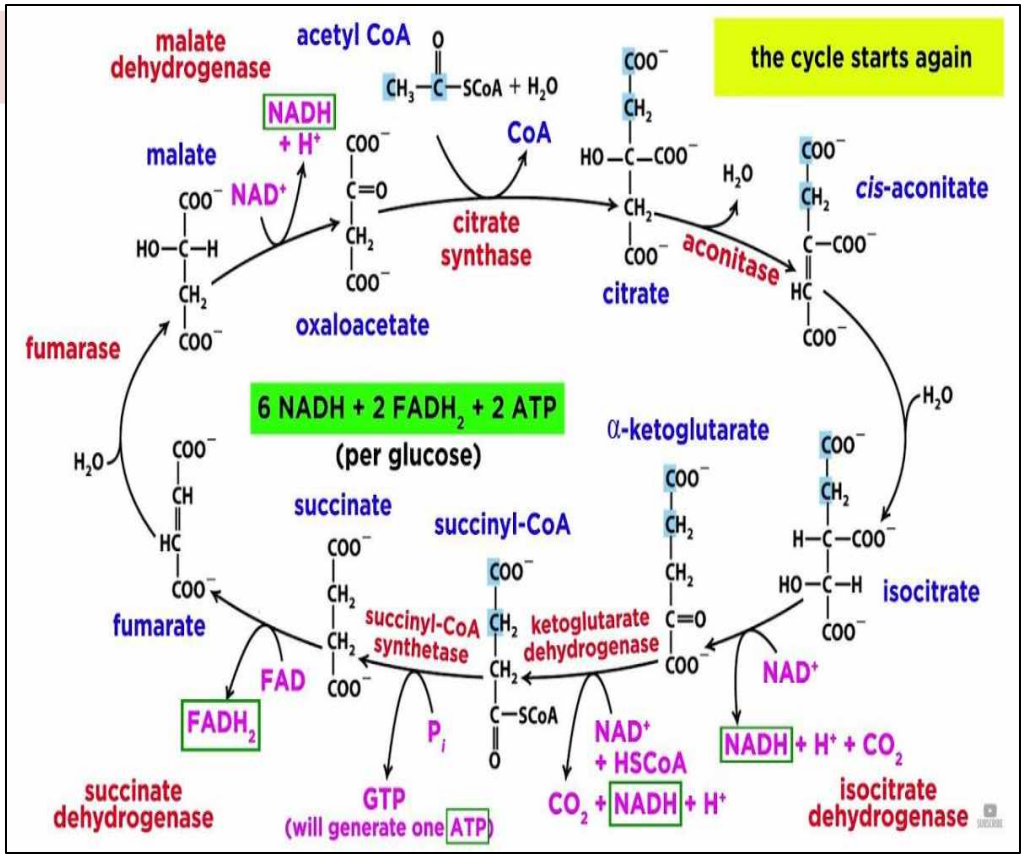
Thus, TCA cycle is central to energy generation from cellular respiration.

- ✓ Acetyl CoA is oxidized to **Two molecules of CO2** and the energy is conserved as **NADH, FADH2, and GTP**
- ✓ **NADH and FADH2** subsequently donate electrons to O2 via the **electron transport chain**, with the generation of ATP **(oxidative phosphorylation)**.

I. The first step, condensation of **acetyl CoA** and **oxaloacetate** to form **citrate** catalyzed by ***Citrate Synthase***

II. **Citrate** is modified to **isocitrate** by ***Aconitase*** enzyme.





III. **Isocitrate** undergoes **oxidative decarboxylation** by ***Isocitrate Dehydrogenase*** to form **α-ketoglutarate.** ***Isoitrate dehydrogenase*** is **NAD**-dependent enzyme. This reaction leads to release of the **first CO2** and formation of **NADH**.

IV. **α-ketoglutarate** undergoes **oxidative decarboxylation** catalyzed by ***α-Ketoglutarate Dehydrogenase Complex***. **Succinyl CoA** is formed.

- ✓ This reaction is very similar to the reaction catalyzed by the **PDH (pre-Krebs).**
- ✓ ***α-ketoglutarate dehydrogenase complex*** is a complex formed of **Three enzymes** bound together (***α-ketoglutarate dehydrogenase, dihyrolipoyl transacetylase*** and ***dihydrolipoyl dehydrogenase***) and **Five coenzymes (Thiamine pyrophosphate (TPP), Lipoic acid, FAD and NAD, Coenzyme A).**
- ✓ ***α-ketoglutarate dehydrogenase complex*** leads to release of the **second CO2**, and formation of **NADH.**
- ✓ However, α-ketoglutarate dehydrogenase **does not subjected to covalent modification by hormones** like PDH.
- ✓ This reaction is **inhibited by *Arsenite*** (an inhibitor of lipoic acid-requiring enzymes).

V. Succinyl CoA is converted to **succinate** by **Succinate Thiokinase** (or **Succinyl CoA Synthetase**).

- ✓ مهم جداااااThis reaction is an example of **substrate level phosphorylation.**

*substrate level phosphorylation is the formation of a high-energy phosphate bond (ATP or GTP) **without** the use of molecular **O2 or the ETC** (it is NOT an oxidative phosphorylation).*

- ✓ The high-energy phosphate bond of GTP is energetically **equivalent** to that of ATP.

VI. Succinate is oxidized by ***Succinate Dehydrogenase*** (**a part of inner mitochondrial membrane**) to form **fumarate**.

- ✓ مهم جداااا**This is the only step in TCA cycle where FAD is the cofactor for the oxidation reaction**.
- ✓ **Succinate dehydrogenase** is competitively **Inhibited** by **Malonate**.

VII. Fumarate by the action ***fumarase*** is hydrated leading to the formation of **malate.**

VIII. Malate is converted to **oxaloacetate**, to close the cycle. The reaction is catalyzed by the ***Malate Dehydrogenase*** enzyme and **NADH** is produced.

To summarize: in the citric acid cycle, the two carbons of **acetyl CoA** are oxidized and released in the form of **2 CO2 + 3 NADH+ one FADH2 +one GTP.**

ENERGY YIELD OF TCA

Each **NADH** will produce **3 ATP** ⟶ **3 NADH X 3= 9 ATP**
Each **FADH2** will produce **2 ATP** ⟶ 1 **FADH2 X 2= 2 ATP**
} **By oxidative phosphorylation In ETC**

An extra **ATP (GTP)** is produced in the cycle by **the substrate level phosphorylation** .

سؤال مهمthe final yield of energy per molecule of **Acetyl CoA** oxidized in the TCA cycle equals **12 ATP**.

سؤال مهمم**In which reactions**

- The **complete oxidation** of one molecule of **Glucose** under **Aerobic conditions** will yield **36-38 ATP** as follows:

Aerobic glycolysis (**6-8 ATP**) according to shuttle
Oxidative decarboxylation of 2 pyruvates (**6 ATP**) by **PDH**
Krebs cycle for 2 molecules of acetyl CoA (**12 X 2 = 24 ATP**)

- **Oxidation of Glucose under anaerobic conditions will only yield 2 ATP.**

N.B **Four members** of the vitamin B-complex (in the form of their coenzymes) play an important role in the TCA cycle
1- Niacin (vit B3) in the form of **NAD+**
2-Riboflavin (vit B2) in the form of **FAD**
3-Thiamin (vitamin B1) in the form of **TPP**
4-pantothenic acid as a part of **coenzyme A (CoASH)**

Note: All the coenzymes of the TCA cycle, except for lipoic acid, are derived from vitamins

THE IMPORTANCE AND THE AMPHIBOLIC NATURE OF THE TCA

Although Krebs cycle is mainly **Catabolic**, it has some **Anabolic** features (It is an **Amphibolic cycle)**.

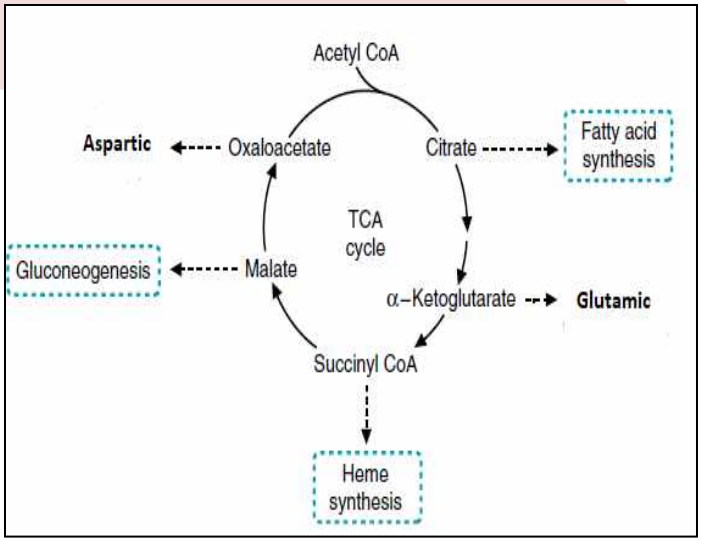
I- The Catabolic nature

TCA cycle is **the final common Catabolic pathway** for oxidation of metabolic intermediates derived from **carbohydrates, lipids and proteins.**

II- The Anabolic nature

-Some intermediates of TCA cycle participate in synthesis of other important biomolecules in different pathways in different cell types.

❖ Anabolic intermediates:



(1) After a high carbohydrate meal, **Citrate** **efflux** from mitochondria and cleavage to **Acetyl CoA** for cytosolic **Fatty acid synthesis** (lipogenesis).

(2) During fasting, gluconeogenic precursors are converted to **Malate**, which leaves the mitochondria for cytosolic part of **gluconeogenesis.**

(3) The liver also uses TCA cycle intermediates to synthesize **Non-essential Amino Acids**. For example:

- **α-ketoglutarate** is converted to **glutamate**
- **oxaloacetate** to **aspartate**
- **Pyruvate** to **alanine**

} **Transamination reactions.**

(1) **Succinyl CoA** can be utilized to form **Heme** in cells of the liver and bone marrow.

Important notes

- **All** the enzymes are in the **matrix** of the mitochondria except **succinate dehydrogenase**, which is in the inner mitochondrial membrane.
- **Succinate thiokinase** (**Succinyl CoA synthetase**) catalyzes a **substrate-level phosphorylation** of GDP to GTP.
- Deficiency of any of the enzymes of the TCA has not been reported, since it is incompatible with life.

REGULATION OF THE TCA CYCLE

1. Coarse control

- ✓ The **main function** of the TCA cycle is **energy production**, which depends mainly on **electron transport chain**.
- ✓ **Coarse control** depends on availability of **oxygen** and the **concentration of ADP**.
- ✓ So, the rate of **utilization of ATP** determines both the rate of respiration and the activity of the TCA cycle.

2. Fine control: Allosteric Regulation

❖ **Regulatory steps are:**

Citrate synthase, i***socitrate dehydrogenase***, ***α-ketoglutarate dehydrogenase***

This is achieved at special steps in the cycle:

a) The ***Citrate synthase***, i***socitrate dehydrogenase***, ***α-ketoglutarate dehydrogenase*** are all activated by **Ca^{++}**, which is released during muscle contraction.

b) All the ***dehydrogenases*** (i***socitrate dehydrogenase***, ***α-ketoglutarate dehydrogenase***)in the cycle are **activated** by low energy level in the cell **(high ADP/ATP, high NAD/NADH),** and vice versa.

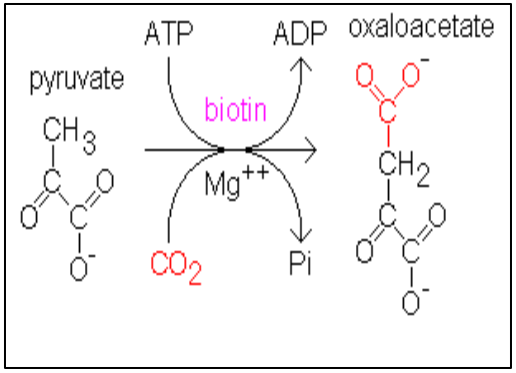
c) High concentration of ATP, long chain fatty acyl CoA and citrate allosterically inhibit ***citrate synthase (feedback inhibition).***

N.B All the ***dehydrogenases*** in the cycle are **inhibited** by high energy level in the cell **(high ATP, high NADH)**

ANAPLORETIC REACTIONS
THAT REPLENISH ("FILLING UP") THE INTERMEDIATES OF THE TCA CYCLE

✓ Since many intermediates of TCA cycle can be used for biosynthesis of some important derivatives so you need to replenish the intermediates of TCA cycle by reactions which are called anaploretic reactions ("filling up") esp Oxaloacetate.

PYRUVATE CARBOXYLASE



- Pyruvate carboxylase is one of the major anaploretic enzymes in the cell.
- It catalyzes the addition of CO2 to **pyruvate** to form **oxaloacetate**.
- found in many tissues, such as liver, brain, adipocytes, and fibroblasts.
- Like most **carboxylases**, pyruvate carboxylase contains **biotin, ATP**
- Pyruvate carboxylase is found in many tissues, such as liver, brain, adipocytes, and fibroblasts.

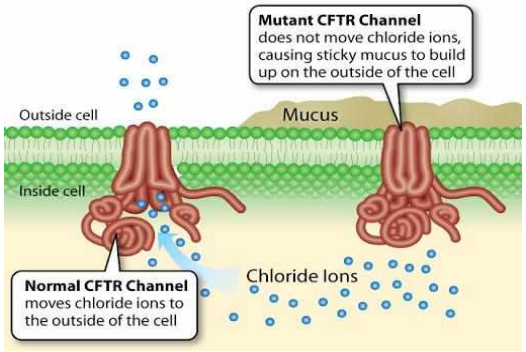
N.B Pyruvate carboxylase is **activated** by **acetyl CoA** (When acetyl CoA accumulates, it allosterically stimulates ***pyruvate caboxylase*** to ensure proper concentration of oxaloacetate for the oxidation of the excess acetyl CoA).

TRANSAMINATION OF NON-ESSENTIAL AMINO ACIDS

1. **α-ketoglutarate** is formed from **glutamate**
2. **oxaloacetate** from **aspartate** (through **transamination reactions**).

MOLECULAR BASIS OF LUNG DISEASES

1. CYSTIC FIBROSIS:



- ✓ **Cystic fibrosis** is due to **deletion of the codon of phenylalanine** in the gene of **CFTR** protein **(Tri-nucleotide deletion).**
- ✓ **CFTR** (cystic fibrosis trans-membrane regulator) acts as a **chloride channel** in epithelial cells, and its loss results in the production of thick, sticky secretions in the lungs and pancreas
- ✓ It is the most common serious genetic disease in the UK.
  - It affects **the lungs** where **thick mucus builds up causing repeated chest infections** and breathing difficulties
  - And the **pancreas** **where blockages cause digestive problems.**
- ✓ The condition is life threatening. Death occurs usually by the 2nd or 3rd decades**.**

2. THE α 1 ANTITRYPSIN DEFICIENCY (EMPHYSEMA)

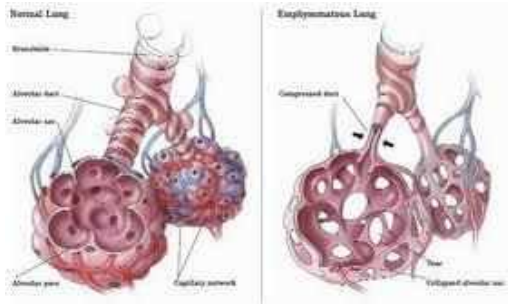
Role of α1 antitrypsin in protection of elastin in the lungs

- **Elastin** presents in the **lung** and is protected from degradation by **elastase effect** by **α1 anti-trypsin** which inactivate elastase
- If there is **α1 anti-trypsin deficiency** → **Increase elastase activity** → **Increase degradation of elastin**

→ **Lung emphysema** (no elastic recoil)

Destruction in connective tissue of alveolar wall which can't be regenerated

- ✓ **α1- antitrypsin** is synthesized by **liver**, **alveolar monocytes**, **macrophages.**
- ✓ **α1- antitrypsin deficiency** is a genetic defect alters the configuration of the alpha1-antitrypsin molecule and prevents its release from hepatocytes & decreased serum levels of alpha1-antitrypsin and low alveolar concentrations

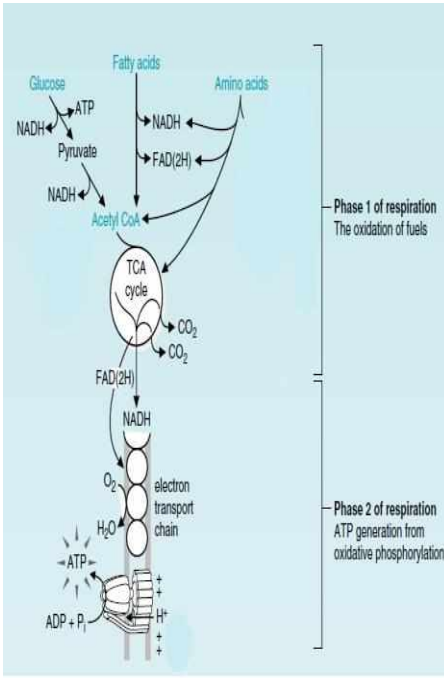


N.B accumulation of mutated alpha1-antitrypsin in hepatocytes ⟶ destruction of hepatocytes and ***liver disease***.

N.B. **Smoking** adds to the severity of the disease.
Smoking causes **oxidation** and **inactivation** of **methionine** in **α1- antitrypsin** (Methionine 358 in α1- antitrypsin is required for the binding of α1- antitrypsin to its elastase).
So, α1- antitrypsin can't bind to elastase. This leads to lung emphysema.

OXIDATIVE PHOSPHORYLATION

❖ Overview of cellular respiration



- ✓ Most of the **energy from oxidation of fuels** in the **TCA cycle** and other pathways is **conserved** in the form of NADH and $FADH_2$.
- ✓ Energy from fuel oxidation is converted to the high-energy phosphate bonds of adenosine triphosphate (ATP) by the process of **oxidative phosphorylation.**
- ✓ Through **ETC, NADH** and **$FADH_2$ are oxidized**, and the electrons are donated to **O_2**, forming H_2O
- ✓ Part of the released energy during transfer of electrons through the ETC is captured for the phosphorylation of adenosine diphosphate (ADP) to ATP by **ATP synthase (F0 F1 ATPase**).
- ✓ The net yield of oxidative phosphorylation is approximately **3 ATP moles of ATP** per mole of **NADH** oxidized, or **2 moles of ATP** per mole of **$FADH_2$** oxidized.

❖ Important definitions

- **Oxidation:** It is the addition of oxygen, or removal of hydrogen, or removal of electrons.
- **Reduction** is the addition of hydrogen, or addition of electrons. or removal of oxygen.

NB. Oxidation of a compound is always **accompanied** by the **reduction** of another compound. This is why we usually refer to the reactions as **oxidation-reduction** reactions.

- **Oxidoreductases:** are enzymes concerned with the catalysis of oxidation-reduction reactions. They are subdivided into **4 subgroups**:

1- Oxidases:

-Enzymes concerned with **removal of hydrogen**, using **oxygen** as a **hydrogen acceptor**.

-They form **water** or hydrogen peroxide.

-An important example of oxidases is the **cytochrome oxidase**. It presents at terminal components of the respiratory chain that is responsible for the reaction with molecular oxygen to produce water.

2- Dehydrogenases:

-They **transfer hydrogen** from one substrate to another, in a **reversible**, coupled oxidation-reduction reaction.

-**Oxygen** is **not used** as a hydrogen acceptor.

-Many dehydrogenases use **nicotinamides** as coenzymes. These may be either **NAD^+ or $NADP^+$.**

-**NAD^+** is usually linked to dehydrogenases responsible for **oxidative pathways** as glycolysis, TCA cycle and the respiratory chain. While, **$NADP^+$** is used by dehydrogenases linked to **reductive synthetic pathways** as in the cytoplasmic synthesis of fatty acids.

-Other dehydrogenases depend on **riboflavin as coenzymes (FAD, FMN)** as succinate dehydrogenase and mitochondrial glycerol 3-phosphate dehydrogenase

3- Hydroperoxidases:

-Two subgroups. They are responsible for **disposal of hydrogen peroxides** from the body (as **peroxidases and catalase)**

4- Oxygenases:

-They catalyse the **incorporation of oxygen** into substrate molecules, with the synthesis or degradation of different metabolites. Have no relation to energy in the body. They are subdivided into two subgroups;

a. Di-oxygenases: they incorporate **both atoms of oxygen molecule** into the substrate (in amino acid metabolism).

$$A + O_2 \longrightarrow AO_2$$

b. Monooxygenase (hydroxylase): These enzymes introduce **one atom of oxygen** to the **substrate** and the **other oxygen atom** is reduced to **water**.

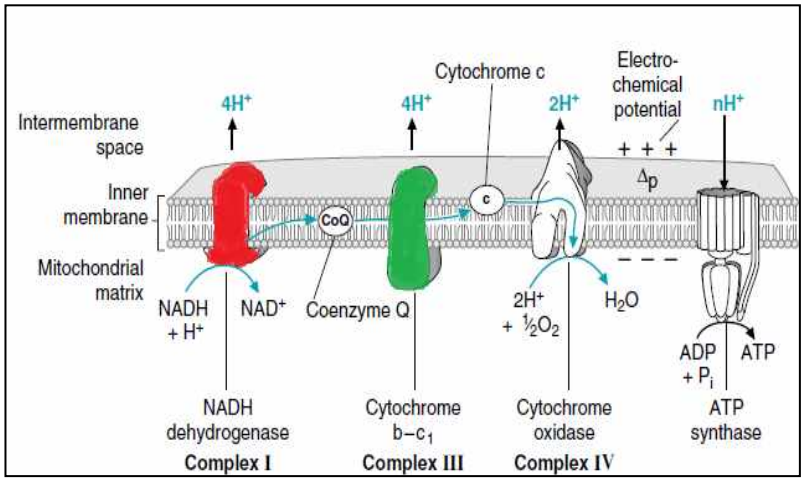
e.g., **Microsomal cytochrome P450** is an important enzyme of this group that is responsible detoxification of many harmful substances and drugs….etc

$$AH + O2 + ZH_2 \longrightarrow A\text{-}OH + H_2O + Z$$

OXIDATIVE PHOSPHORYLATION IN ETC

Generation of ATP by oxidative phosphorylation requires:

- an electron donor (NADH or $FADH_2$)
- an electron acceptor (O_2)
- an intact inner mitochondrial membrane
- All the components of the electron transport chain
- ATP synthase
  - Oxidative phosphorylation is **regulated** by the rate of **ATP utilization**.



❖ **Overview of the electron transport chain (ETC)**

- Electrons donated by NADH or $FADH_2$ are passed sequentially through a series of electron carriers embedded in the **inner** mitochondrial membrane.
- Each of the components of the ETC is **reduced** as it accepts an electron, and then **oxidized** as it passes the electrons to the next member of the chain, **The final acceptor is O_2, which is reduced to H_2O.**
- With the exception of coenzyme Q (CoQ), all members of the ETC are proteins.
- The ETC included 4 complexes: **NADH dehydrogenase (Complex I)**
  Ssuccinate dehydrogenase (Complex II)
  Cytochrome b-c1 complex (Complex III)
  Cytochrome oxidase (Complex IV).
- In oxidation-coupled phosphorylation, the **energy for ATP synthesis** is provided by **an electrochemical gradient** across the **inner mitochondrial** membrane.
- This **electrochemical gradient** is **generated** by the **components of the electron transport chain**, which **pump protons** across the inner mitochondrial membrane as they sequentially **accept and donate electrons**.

- NAD^+ is reduced to NADH by a dehydrogenase (e.g., isocitrate dehydrogenase of TCA cycle) that removes two hydrogen atoms from their substrate.
- Both electrons but only one proton (a hydride ion [:H–]) are transferred to the NAD^+, forming NADH plus a free proton H^+.

Components and Carriers of ETC

1-COENZYME Q:

- Coenzyme Q (CoQ) is a quinone derivative with a long, hydrophobic isoprenoid tail.
- It is also called **ubiquinone** because it is **ubiquitous** in biologic systems.
- **CoQ** is a **mobile** electron carrier and can **accept hydrogen atoms** from NADH dehydrogenase (**Complex I**), from succinate dehydrogenase (**Complex II**) and from other **mitochondrial dehydrogenases**: as glycerol-phosphate dehydrogenase
- **CoQ** transfers electrons to **Complex III** (cytochrome bc1).

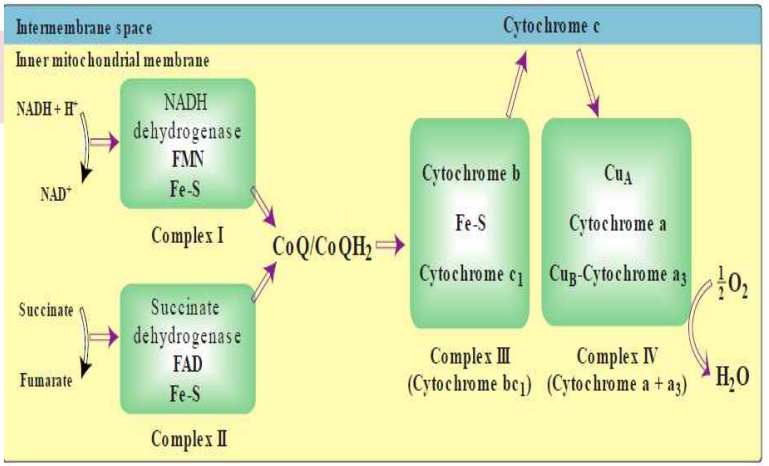
N.B: The free energy drop between **NADH** and **CoQ** is able to pump protons. However, the **$FADH_2$** in succinate dehydrogenase (as well as glycerophosphate dehydrogenase) is at roughly the same energy level as **CoQ**, and there is no energy released as they transfer electrons to CoQ.

2- CYTOCHROMES :

- The remaining members of the ETC are cytochrome proteins, Each contains a **heme** group (a **porphyrin** ring& **iron**).
- Ex: Cytochrome C (the only free carrier), a-a3, b-c1
- **Unlike** the iron of the heme groups of haemoglobin, the cytochrome iron is **reversibly** converted from its **ferric** (Fe^{3+}) to its **ferrous** (Fe^{2+}) form as a normal part of its function as an **acceptor and donor of electrons**.

3- Other electron Carriers: **Iron-sulfur protein, FMN**

1. NADH DEHYDROGENASE (COMPLEX I)



- The **free proton plus the hydride ion** carried by NADH are transferred to NADH dehydrogenase, a protein complex (**Complex I)** that is embedded in the inner mitochondrial membrane.
- **Complex I** has a tightly bound molecule of **flavin mononucleotide** (FMN), a coenzyme structurally related to FAD that accepts the two hydrogen atoms (2e– + 2H+), becoming **$FMNH_2$**. NADH dehydrogenase also contains peptide subunits with **iron-sulfur centers**

- **At Complex I**, electrons move from **NADH** to **FMN** to the **iron of the iron-sulfur centers** and then to **coenzyme Q**.
- As **electrons** flow, they **lose energy**. This energy is used to **pump protons** across the inner mitochondrial membrane, from **the matrix** to the **intermembrane space**.

2. SUCCINATE DEHYDROGENASE (COMPLEX II)

- Succinate dehydrogenase catalyzes the oxidation of succinate to fumarate and the reduction of the coenzyme, **FAD to $FADH_2$**. Then, hydrogen atoms are moved from $FADH_2$ to an **iron-sulfur** protein, and then to **coenzyme Q**. [Note: ***No energy is released in this process, and, therefore, no protons are pumped at Complex II***].

3. CYTOCHROME b-c1 (COMPLEX III),

- Electrons move from **cytochrome b** to **the iron-sulfur centers** then to **cytochrome c1** then to to **cytochrome c**, and then to **cytochromes a + a3** (Complex IV).
- As electrons flow, **protons** are pumped across the inner mitochondrial membrane at Complexes **III and IV**.
- The cytochromes of the **b-c1** complex have a **higher energy level** than those of **cytochrome oxidase (a + a3)**. Thus, energy is released by electron transfer between complexes III and IV.

Note: **Cytochrome c** is located in the intermembrane space, **loosely** associated with the outer face of the inner membrane. As seen with CoQ, cytochrome c is also a mobile carrier of electrons.

4- CYTOCHROME a + a3 (COMPLEX IV)

- This cytochrome complex (**Complex IV**) is the **only** electron carrier in which the heme iron has an available coordination site that can react directly with **O_2** and so also is called **cytochrome oxidase.**
- At Complex IV, the transported electrons, O_2, and free protons are brought together, and O_2 is reduced to water
- **Note: Four electrons are required to reduce one molecule of O_2 to two molecules of water.**
- Cytochrome oxidase contains copper **(Cu) atoms.**
- **Electrons** move from **CuA** to **cytochrome a** to **cytochrome a3** (in association with **CuB**) to **O_2 which is the final acceptor**.

How is oxidation coupled to the production of the high-energy compound ATP?

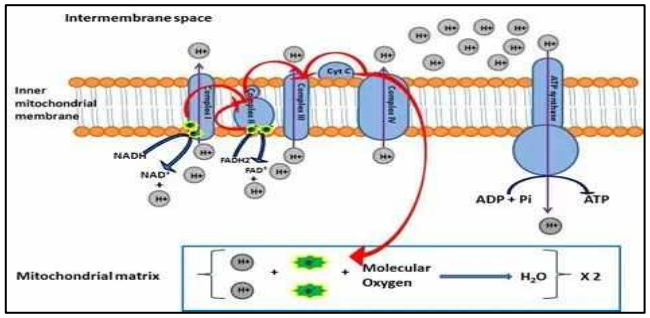
✓ The transfer of electrons down the ETC is energetically favored because **NADH** is a **strong electron donor** and **O_2** is **a strong electron acceptor**. However, the flow of **electrons** does **not** directly result in **ATP synthesis**.

CHEMIOSMOTIC HYPOTHESIS

The chemiosmotic hypothesis (also known as the **Mitchell hypothesis**) explains how the **free energy** generated by the **transport of electrons** by the ETC is used to **produce ATP** from ADP + Pi.

1. Proton pump:

- **Electron transport** is **coupled** to the **phosphorylation** of **ADP** by the pumping of protons across the inner mitochondrial membrane from the **matrix** to the **intermembrane space** at Complexes **I**, **III**, and **IV**.
- This process creates **an electrical gradient** (with more positive charges on the outside of the membrane than on the inside) and a **pH gradient** (the outside of the membrane is at a lower pH than the inside)
- **The energy generated by this proton gradient is sufficient to drive ATP synthesis (Complex V)**
- Thus, the **proton gradient** serves as the **common intermediate** that **couples oxidation to phosphorylation.**



2. ATP synthase (Fo F1ATPase)

- The multi-subunit enzyme ATP synthase (**Complex V**) synthesizes ATP using the energy of the proton gradient.
- It contains a domain **(Fo)** that spans the inner mitochondrial membrane, and an extra-membranous **domain (F1)** that appears as a sphere that protrudes into the mitochondrial matrix
- **The chemiosmotic hypothesis proposes that after protons have been pumped to the cytosolic side of the inner mitochondrial membrane, they re-enter the matrix by passing through a proton channel in the Fo domain, driving rotation of Fo and, at the same time, dissipating the pH and electrical gradients.**

- **Fo rotation** causes **conformational changes** in the β subunits of the **F1** domain that allow them to bind ADP + Pi, **phosphorylate ADP to ATP**, and release ATP.

❖ ***Several findings support the chemiosmotic theory:***

- **Addition of protons** (acid) to the external medium of intact mitochondria stimulates the formation of **ATP.**
- Oxidative phosphorylation **does not occur** in case of **solubilizing** the mitochondrial membranes.

❖ ***Respiratory control ensures a constant supply of ATP***

✓ It is to be noted that there is **no mechanism** for **storage** of ATP in the body.
✓ Actually all the **ATP present at any moment is only enough to meet the needs of the body for a few seconds**.
✓ So, there must be an efficient and controlled way for the production of ATP. The respiratory chain allows the free energy resulting from the oxidation of food stuffs to become available in a stepwise, efficient and controlled way.

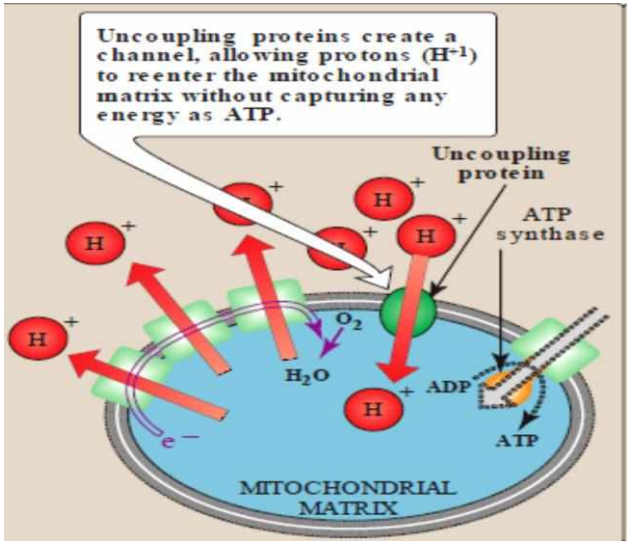
CONTROL OF THE RESPIRATORY CHAIN (REGULATION OF ETC)

- Mainly by the **availability of ADP**, that constitutes the **main** controlling factor.
- When doing work, **ATP** is converted to **ADP**. This allows more **respiration** (oxidation reactions) to **occur**, which in turn **replenishes** the stores of **ATP**.

- Under certain conditions, the **concentration of inorganic phosphate** could affect the rate of ATP formation.

INHIBITORS OF RESPIRATORY CHAIN

1- UNCOUPLERS:



When protons leak back into the mitochondrial matrix **without** going through the **ATP synthase (proton channel in the Fo domain)**, they dissipate the electrochemical gradient across the membrane **without** generating ATP. This phenomenon is called "**uncoupling**" of oxidation to phosphorylation

- It occurs with chemical compounds, known as **uncouplers,** and it may occur physiologically or not, with uncoupling proteins that form proton conductance channels through the membrane.
- Uncoupling of oxidative phosphorylation results in **increased oxygen** consumption and **heat production** as electron flow and proton pumping attempt to maintain the electrochemical gradient.

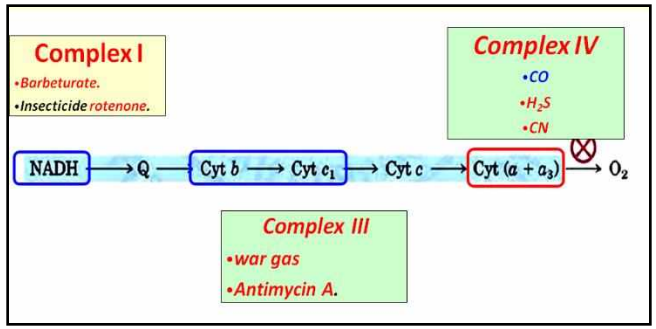
❖ ***Examples of uncouplers:***

- **Thermogenin (UCP1):**
  - It is considered as **a physiological** uncoupling protein.
  - It is present in the **brown adipose tissues** of newly born humans.
  - It allows **protons** to pass to the **mitochondrial matrix** without passing by the **F0-F1 complex**.
  - **No ATP** is produced and **energy is dissipated** in the form of **heat**.
- **Synthetic uncouplers**: **2,4-dinitrophenol**, a lipophilic proton carrier that readily diffuses through the mitochondrial membrane.

2-INHIBITORS OF THE RESPIRATORY CHAIN COMPONENTS

These inhibitors act on the three sites that produce the electrochemical gradient; i.e., complexes I, III and IV.

- **Complex I** is inhibited by**: barbiturates, and the insecticide rotenone.**
- **Complex II** is inhibited by**: malonate**
- **Complex III** is inhibited by**: the war gas and antimycin A.**
- **Complex IV** is inhibited by: **H_2S**, **carbon monoxide** (**CO**) and **cyanide** (**CN**).



3- OLIGOMYCIN

- This drug binds to the **Fo domain of ATP synthase**, closing the **proton** channel and preventing **re-entry of protons into the matrix**, thereby **preventing** phosphorylation of **ADP to ATP**.
- Because the pH and electrical gradients cannot be dissipated in the presence of this drug, electron transport stops because of the difficulty of pumping any more protons against the steep gradients.

INHERITED DEFECTS IN OXIDATIVE PHOSPHORYLATION

- 13 of the approximately 90 polypeptides required for oxidative phosphorylation are **coded for by mitochondrial DNA (mtDNA)** and synthesized in **mitochondria.**
- The **remaining** proteins are coded for by **nuclear DNA**, synthesized in the **cytosol,** and transported into the **mitochondria.**
- **Defects** in oxidative phosphorylation are more likely a result of **alterations in mtDNA,** which has a **mutation rate about 10 times greater** than that of **nuclear** DNA.
- **Tissues** with the greatest ATP requirement (central nervous system, skeletal and heart muscle, and liver) are most **affected** by **defects** in oxidative phosphorylation.

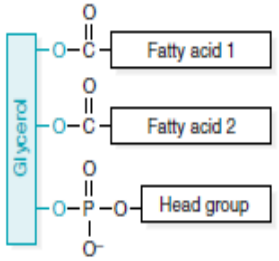
- **Mutations in mtDNA** are responsible for several diseases, including some cases of **mitochondrial myopathies**, and **Leber hereditary optic neuropathy** (a disease in which bilateral loss of central vision occurs as a result of **neuroretinal degeneration, including damage to the optic nerve**).

Note: mtDNA is **maternally-inherited** because **mitochondria** from the **sperm cell <u>do not</u>** enter the fertilized egg.

PHOSPHOLIPIDS METABOLISM

❖ **The chemical structure and types of phospholipids: (in introductory module of biochemistry**

Alcohol, Fatty acid, and phosphoric acid.

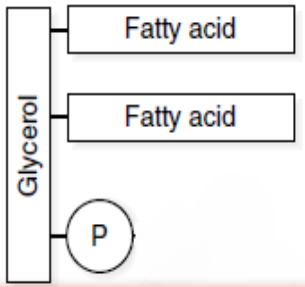


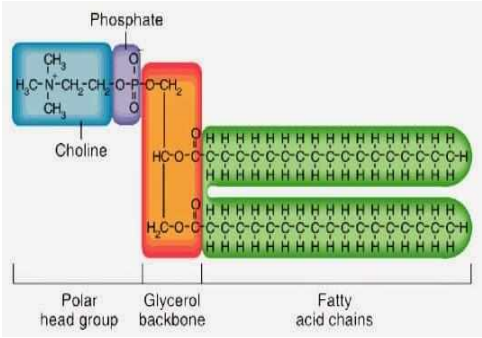
1. <u>Glycero-phospholipids:</u>Glycerol alcohol

1. Phosphatidic acid.**(Parent)**
2. Lecithin (phosphatidylcholine).
3. Cephalin (phosphatidylethanolamine).
4. Phosphatidylinositol
5- Cardiolipin

<u>2- Sphingo phospholipids</u>

- **Sphingomyelin.**





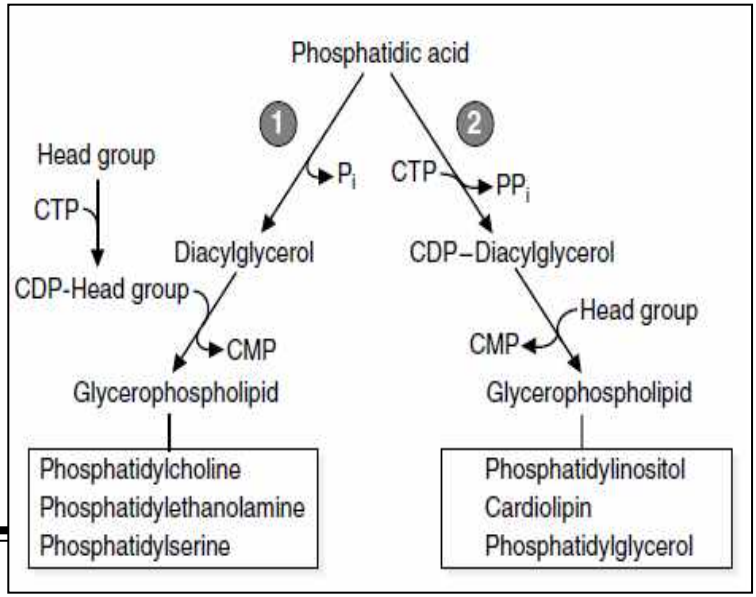
SYNTHESIS OF PHOSPHOLIPIDS CONTAINING GLYCEROL

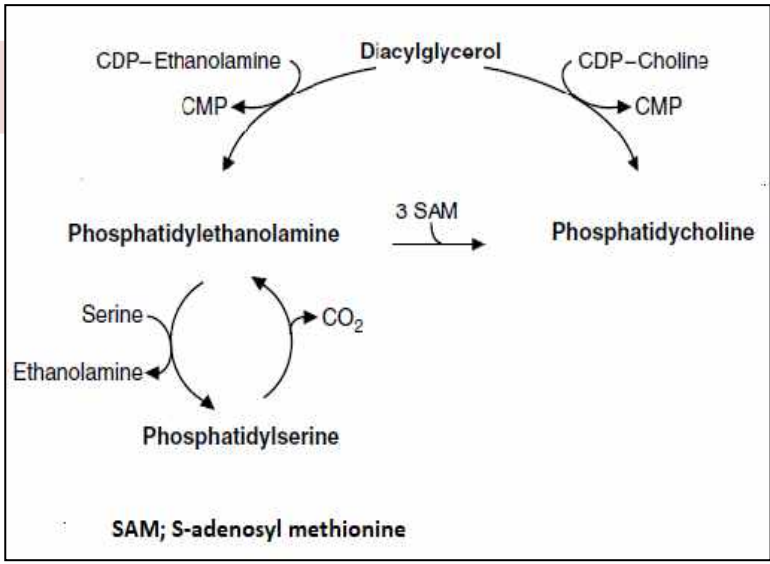
- **Glycerol 3-phosphate** reacts with **fatty acyl CoA** to form **phosphatidic acid**.
- **Two different pathways to add a head group to the molecule**

❖ **<u>In the first pathway:</u>**

Phosphatidic acid is cleaved by a **phosphatase** to form **diacylglycerol** (DAG).

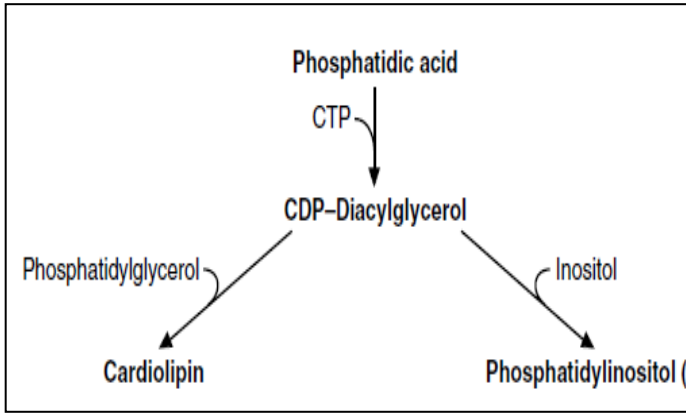
- **DAG** then reacts with an **activated head group**.





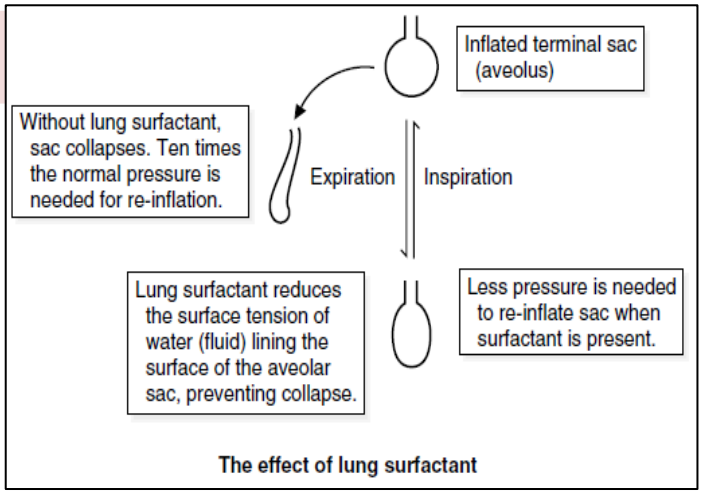
- In the synthesis of **phosphatidylcholine**, the head group **choline** is activated by combining with **CTP** to form **CDP-choline**. **Phosphocholine** is then transferred to **carbon 3** of **DAG**, and **CMP** is released.
- **Phosphatidylethanolamine** is produced by a similar reaction involving CDP-ethanolamine.
- **Phosphatidylserine** is produced by a reaction in which the **ethanolamine** moiety of phosphatidylethanolamine is **exchanged for serine**.
- **Phosphatidylserine** can be converted back to **phosphatidylethanolamine** by a **decarboxylation reaction**.
- **Phosphatidylethanolamine** can be **methylated** to form **phosphatidylcholine (through 3 methylation steps)**. S-adenosyl methionine (SAM) is the methyl donor.

❖ **In the second mechanism:**



- **Phosphatidic acid** reacts with **CTP** to form **CDP-diacylglycerol**.
- **CDP-diacylglycerol** reacts either with **phosphatidylglycerol** (which itself is formed from the condensation of CDP-diacylglycerol and glycerol 3-phosphate) to produce **cardiolipin** or with **inositol** to produce **phosphatidylinositol.**
- **Phosphatidylinositol** can be **phosphorylated** to form **phosphatidylinositol 4,5-bisphosphate** (PIP2), which is a **component of cell membranes**. In response to **signals** such as the **binding of hormones to membrane receptors**, PIP2 can be cleaved to form the second messengers: diacylglycerol and inositol tri-phosphate.

Clinical significance:



The effect of lung surfactant

Respiratory distress syndrome (RDS):

Neonatal defect in synthesis of the **phospholipids Dipalmitoyl Phosphatidyl choline (Dipalmitoyl lecithin, Lung surfactant)**

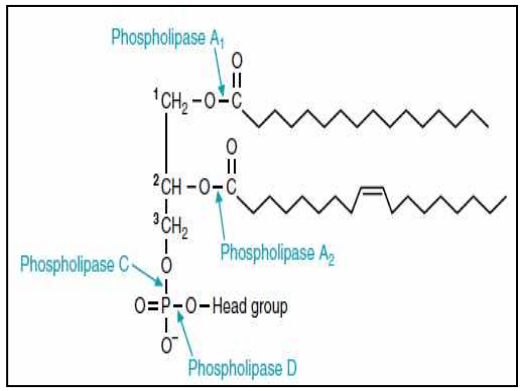
Lung surfactant **decreases** the **surface tension** in the lung alveoli. So, it **prevents adherence** of the inner surfaces of the lung at the **end of expiration**

Neonatal respiratory distress syndrome (RDS):

- RDS in premature infants is associated with **insufficient surfactant production** and/ or **secretion**.
- RDS is preventable if prematurity can be avoided by appropriate management of high-risk pregnancy.
- **Lung prematurity** of the fetus determined by the ratio concentrations of **phosphatidylcholine (lecithin)** to **sphingomyelin** in the amniotic fluid (value of 2 or more indicates lung maturity).
- The administration of synthetic **corticosteroids 48 to 72 hours before delivery** of a fetus of less than 33 weeks of gestation in women who have toxemia of pregnancy, diabetes mellitus, or chronic renal disease **may reduce the incidence or mortality of RDS by stimulating fetal synthesis of lung surfactant**.
- The administration of **one dose of surfactant into the trachea of the premature infant immediately after birth** may transiently improve respiratory function but does not improve overall mortality.

DEGRADATION OF GLYCEROPHOSPHOLIPIDS

➢ **Phospholipases** located in **cell membranes** or in **lysosomes** degrade **glycerophospholipids**.



- **Phospholipase A1** removes the **fatty acyl** group on **carbon 1** of the glycerol moiety.
- **Phospholipase A2** removes the **fatty acid** on **carbon 2**. The fatty acid attached to C2 in cell membrane phospholipids is usually an **unsaturated fatty** acid, which is frequently **arachidonic acid**. It is removed in response to signals for the **synthesis of eicosanoids**. Phospholipase A2 is inhibited by glucocorticoids, e.g., cortisol.
- The bond joining **carbon 3** of the glycerol moiety to **phosphate** is cleaved by **phospholipase C**.
- Hormonal stimuli activate phospholipase C, which hydrolyzes PIP2 to produce the second messengers DAG and inositol trisphosphate (IP3).
- The bond between the phosphate and the head group is cleaved by **phospholipase D**, producing **phosphatidic acid** and the **free alcohol of the head group**.

❖ Phospholipidosis

Lysosomal storage disorder characterized by excess accumulation of phospholipids in tissues (Kidney, lung, liver).

- It may be **inherited or acquired.**
- In acquired phospholipidosis, some drugs bind to phospholipids and inhibit their hydrolysis by lysosomal phospholipases. This condition is reversible after discontinuation of drug treatment.